

Basket Ball By The Numbers

Project Report

Team 4

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Abstract

This project assesses player and team-level performance in the NBA for the 2024–25 season and examines team trends over the past ten seasons. Data were scraped from Basketball Reference using R (rvest, tidyverse) and visualized through interactive dashboards for age–position analysis and team performance trends.

1 Overview

Our project analyses player and team-level performance in NBA matches for the 2024–25 season and studies team-level trends across the previous ten seasons (2015–16 to 2024–25). The main objectives are:

- To characterise player age distribution by position.
- To enable interactive exploration of team metrics across seasons.
- To provide summary statistics for comparing team consistency.

2 Data Sources & Collection

Primary source: basketball-reference.com. Data collected:

- Player per-game statistics for 2024–25 (PTS, MP, FG, FGA, Age, Pos, Team, etc.).
- Team ratings for 2024–25 and historical team ratings (2015–16 to 2023–24) with metrics like ORtg, DRtg, NRTg, Adj_NRTg, MOV, and W/L%.

2.1 Web-Scraping (R)

The R script employs rvest and tidyverse to scrape, clean, and export NBA statistics. Player data are sorted by points per game, while team ratings are structured for adjusted net rating comparison.

```
1 library(rvest); library(tidyverse)
2
3 html = read_html("https://www.basketball-
4 reference.com/leagues/NBA_2025_per_game.html")
5
6 table = html %>% html_table()
7 Player_database = table[[1]]
8 Player_database = Player_database[order(Player_
9 database$PTS, decreasing = TRUE),]
10 Player_database$PTS = as.double(Player_database$
11 PTS)
12 Player_database$Player = as.character(Player_
13 database$Player)
14 write.csv(Player_database, "Player_database.csv")
```

Listing 1: Player scraping (excerpt)

```
1 html = read_html("https://www.basketball-
2 reference.com/leagues/NBA_2025_ratings.html")
3 table = html %>% html_table()
4 team_rating_24_25 = table[[1]][-1,]
5 colnames(team_rating_24_25) = c("Rk", "Team", "Conf",
6 "Div", "W", "L", "W/L%", "MOV", "ORtg", "DRtg", "NRTg",
7 "Adj_MOV", "Adj_ORtg", "Adj_DRtg", "Adj_NRTg")
8 team_rating_24_25 <- team_rating_24_25 %>%
9 mutate(across(c(Rk, W, L, 'W/L%', MOV, ORtg, DRtg,
10 NRTg, Adj_MOV, Adj_ORtg, Adj_DRtg, Adj_NRTg),
11 as.numeric))
12 write.csv(team_rating_24_25, "team_rating_24_25.csv")
```

Listing 2: Team ratings scraping (excerpt)

Rk	Team	Conf	Div	W	L	W/L%	MOV	ORtg	DRtg
1	Shai Gilgeous-Alexander	26	OKC	PG	76	76	34.2	11.3	21
2	Giannis Antetokounmpo	30	MIL	PF	67	67	34.2	11.8	19
3	Nikola Jokic	29	DEN	C	70	70	36.7	11.2	19
4	Luka Dončić	25	2TM	PG	50	50	35.4	9.2	20
4	Luka Dončić	25	LAL	PG	28	28	35.1	8.8	20
4	Luka Dončić	25	DAL	PG	22	22	35.7	9.8	21
5	Anthony Edwards	23	MIN	SG	79	79	36.3	9.1	20
6	Jayson Tatum	26	BOS	PF	72	72	36.4	9.2	20
7	Kevin Durant	36	PHO	PF	62	62	36.5	9.5	18
8	Tyrese Maxey	24	PHI	PG	52	52	37.7	9.2	21
9	Cade Cunningham	23	DET	PG	70	70	35.0	9.8	20
10	Jalen Brunson	28	NYK	PG	65	65	35.4	9.0	18

Rk	Team	Conf	Div	W	L	W/L%	MOV	ORtg	DRtg
1	Oklahoma City Thunder	W	NW	68	14	0.829	12.87	120.33	107
2	Cleveland Cavaliers	E	C	64	18	0.780	9.54	121.88	112
3	Boston Celtics	E	A	61	21	0.744	9.11	120.83	111
4	Minnesota Timberwolves	W	NW	49	33	0.598	5.00	116.72	111
5	Los Angeles Clippers	W	P	50	32	0.610	4.66	115.20	110
6	Houston Rockets	W	SW	52	30	0.634	4.51	115.23	110
7	Memphis Grizzlies	W	SW	48	34	0.585	4.85	117.83	113
8	Denver Nuggets	W	NW	50	32	0.610	3.89	119.97	116
9	New York Knicks	E	A	51	31	0.622	4.10	118.42	114
10	Golden State Warriors	W	P	48	34	0.585	3.30	115.09	111

Figure 1: Screenshots of scraped tables. Top: Player-level data. Bottom: Team ratings for the 2024–25 season.

3 Analysis of Player Age and Positional Specialization

We analysed how player age relates to positional roles using both absolute counts and percentage distributions.

3.1 Key Observations

1. **Peak Age Concentration.** Most NBA players fall within 22–27 years, the typical athletic prime. The

- number drops notably after 30.
- 2. **Positional Longevity.** Centers (C) and Power Forwards (PF) are more common among older players, emphasizing experience and strength. SG and SF dominate younger groups, reflecting agility and speed.
 - 3. **Balanced Representation.** Between 22–27 years, all positions are evenly represented, showing team balance during peak years.

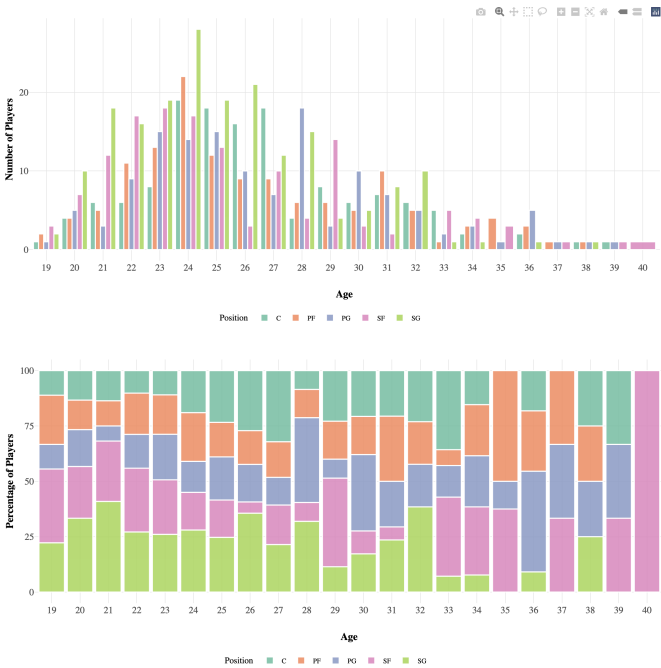


Figure 2: Top: Player counts by age and position. Bottom: Percentage share across age groups.

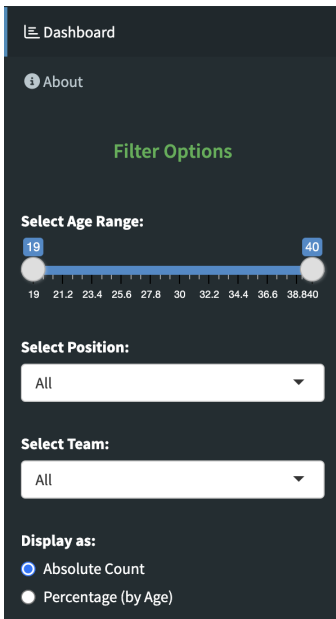


Figure 3: An interactive filter panel lets users select specific age ranges, positions, and teams, and view results as absolute counts or percentage distributions.

4 Analysis of Team Level Performance Metrics

A Shiny dashboard was built to interactively explore team performance metrics.

- Users can:
- Choose any performance metric (e.g., NRtg).
 - Select up to six teams.
 - Adjust season number via slider.

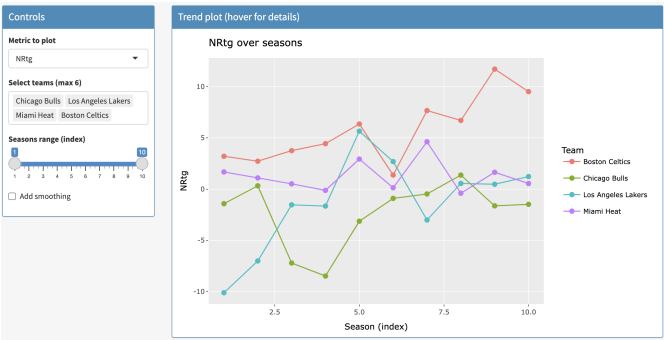


Figure 4: Trend plot showing NRtg variation across seasons.

This plot shows how the selected metric varies over seasons for the chosen teams, with tooltips for detailed values on hover. It helps users visually analyze and compare team performance trends across seasons for chosen metrics.

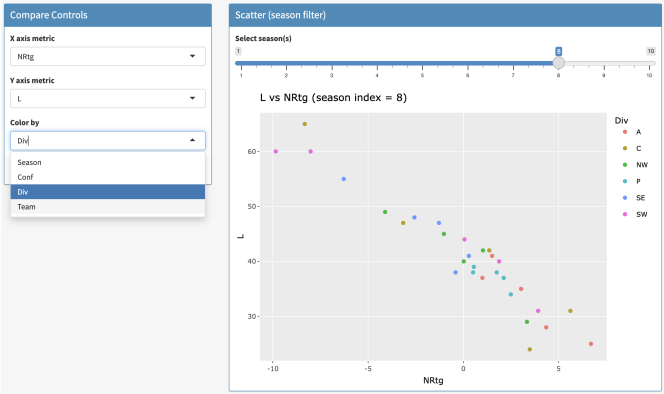


Figure 5: Scatter plot (NRtg vs L) colored by Division.

Users can choose any metrics for the X and Y axes (like NRtg and L) and color the points by attributes such as Division, Conference, Season, or Team.

This Scatter plot helps in checking whether two metrics are directly or inversely correlated for NBA teams. Users can choose different metrics and color categories to better understand performance relationships.

We have also included a summary table that shows the mean and standard deviation of each team’s performance across 10 seasons for a selected metric, helping users compare overall consistency and long-term trends among teams.

5 Exploratory Data Analysis

NBA Player Points Prediction – Multiple Linear Regression Report

Dataset: NBA player performance, 2024–25 season.

Observations: 730 (after cleaning).

Response: PTS (points per 100 team possessions).

Predictors: 18 numeric predictors + 1 categorical predictor (Pos with levels C/PF/SF/PG/SG).

1. Introduction

The dataset contains statistics for 736 NBA players in the 2024–25 season. After removing incomplete rows and irrelevant columns, 730 observations remained. The objective of this analysis is to predict points per 100 team possessions (PTS) using player performance metrics, including age, minutes played, shooting percentages, rebounds, assists, turnovers, and player position.

Variables (detailed)

Below we list each variable included in the final modelling dataset, with data type, unit / scale, how it was computed (if derived), handling of missing values, and modelling notes.

Age

Type: Numeric (continuous).

Units: Years.

Computation / notes: Age at the time of the season. Used as-is; no transformation applied. May capture experience and physical decline effects (nonlinearities possible).

G

Type: Integer (count).

Units: Games played during the season.

Computation / notes: Used to control for availability; high G increases exposure/opportunity. Included as a linear predictor, but could interact with MP or GS.

GS

Type: Integer (count).

Units: Games started.

Computation / notes: Captures role (starter vs bench). In the model GS sometimes shows a small negative coefficient after controlling for MP, suggesting that starts alone do not fully explain scoring once minutes are accounted for.

MP

Type: Numeric (continuous).

Units: Minutes per 100 team possessions (or minutes per game normalized to per-100 possessions depending on scraping); effectively a usage proxy.

Computation / notes: One of the strongest predictors—more minutes directly increase scoring opportunities; treated as continuous and not transformed in primary models. Check for leverage from extremely high MP players.

X3P_perc

Type: Numeric (percentage in %).

Units: 3-point makes per 3-point attempts $\times 100$.

Computation / notes: Derived as $X3P / X3PA * 100$. Missing values (zero attempts) were set to 0 and the percentage recomputed to avoid NaNs. Positively associated with PTS—higher 3P% raises scoring efficiency.

X2P_perc

Type: Numeric (percentage in %).

Units: 2-point makes per 2-point attempts $\times 100$.

Computation / notes: Derived similarly to X3P_perc; missing attempts handled by filling with 0. Highly correlated with X3P_perc for some players—check VIF.

FT_perc

Type: Numeric (percentage in %).

Units: Free-throw makes per free-throw attempts $\times 100$.

Computation / notes: Derived from FT / FTA . Recalculated after filling missing counts with 0. Positively contributes to PTS and often significant in models.

ORB

Type: Numeric (continuous).

Units: Offensive rebounds per 100 team possessions.

Computation / notes: Raw ORB included; tends to be collinear with TRB and DRB. Consider combining or removing to reduce multicollinearity if necessary.

DRB

Type: Numeric (continuous).

Units: Defensive rebounds per 100 team possessions.

Computation / notes: As above; DRB plus ORB yields TRB. High collinearity observed with ORB/TRB (VIF warnings).

TRB

Type: Numeric (continuous).

Units: Total rebounds (ORB + DRB) per 100 team possessions.

Computation / notes: TRB is derived; if retained alongside ORB/DRB expect multicollinearity—use either TRB or the two components.

AST

Type: Numeric (continuous).

Units: Assists per 100 team possessions.

Computation / notes: Proxy for playmaking and primary ball-handler role. May interact with position (PG).

STL

Type: Numeric (continuous).

Units: Steals per 100 team possessions.

Computation / notes: Defensive counting stat; low absolute values but useful to capture defensive activity.

BLK

Type: Numeric (continuous).

Units: Blocks per 100 team possessions.

Computation / notes: Defensive proxy, often higher for bigs (C/PF).

TOV

Type: Numeric (continuous).

Units: Turnovers per 100 team possessions.

Computation / notes: Surprisingly enters the model with a positive coefficient; likely because TOV is correlated with usage rate—players who handle/attempt more also turn over more and also score more.

PF

Type: Numeric (continuous).

Units: Personal fouls per 100 team possessions.

Computation / notes: Can correlate with physical style and minutes; included as control variable.

PTS

Type: Numeric (continuous) — **Response variable.**

Units: Points scored per 100 team possessions.

Computation / notes: This is the target for prediction. For modelling we fit both the raw PTS (linear model) and $\log(\text{PTS} + 1)$ (to stabilize variance).

Pos

Type: Categorical (factor).

Levels: {C, PF, SF, SG, PG}.

Computation / notes: Position encoding: converted to factor, with baseline set to C (center) in regression summaries shown. Position captures role differences in scoring, shot selection, and minutes. Consider combining rare multi-position labels if present (e.g., PF/C).

Missing value policy and derivations:

- For raw shooting counts (FG, FGA, X3P, X3PA, X2P, X2PA, FT, FTA) missing counts were imputed as 0 when the player had no attempts logged, then percentages were recalculated (e.g., $\text{X3P_perc} = \text{X3P} / \text{X3PA} * 100$) to avoid NaNs from zero division.
- Observations with missing critical metadata (e.g., Pos or Age) were removed; final cleaned dataset contains 730 observations.
- All per-game metrics were scaled to *per-100 possessions* to make players comparable across different team tempos. If your source already provided per-100 values, we used them directly.

Notes on multicollinearity and preprocessing:

- Rebound variables (ORB, DRB, TRB) are strongly collinear. For stable coefficient estimates, consider:
 - Removing ORB and DRB and keeping TRB, or
 - Using principal components or regularization (Ridge/Lasso).
- Shooting percentages can correlate (players who take more 3s may have correlated 2P% patterns). Check VIF and consider centering/scaling predictors.
- MP is a dominant predictor — consider including an

interaction term (e.g., MP:Pos) if you hypothesize minutes affect positions differently.

2. Data Cleaning and Preprocessing

- Excluded columns: X, Rk, Player, Team, Awards, and raw shooting counts where unnecessary.
- Missing values in FG/X3P/X2P/FT were replaced with 0 and shooting percentages were recalculated to avoid division by zero.
- Feature engineering: derived X3P_perc, X2P_perc, FT_perc.
- Pos converted to a factor with levels PF, PG, SF, SG, C.
- Final dataset: 18 numeric predictors + 1 categorical predictor + response variable.

3. Model Specification

The full linear model fitted:

$$\text{PTS} = \beta_0 + \beta_1 \text{Age} + \beta_2 \text{G} + \beta_3 \text{GS} + \beta_4 \text{MP} + \dots + \beta_{19} \text{Pos} + \epsilon$$

A *log-transformed* model using $\log(\text{PTS} + 1)$ was also fitted to address heteroscedasticity and improve model assumptions.

4. Regression Results

4.1 Full Model (Untransformed)

- Multiple R^2 : 0.874
- Adjusted R^2 : 0.871
- Significant predictors ($p < 0.05$): G, GS, MP, X3P_perc, X2P_perc, FT_perc, TOV, PF, PosPG (relative to baseline Pos=C)

```
Call:
lm(formula = PTS ~ Age + G + GS + MP + X3P_perc + X2P_perc +
    FT_perc + ORB + DRB + TRB + AST + STL + BLK + TOV + PF +
    Pos, data = player_data_cleaned)

Residuals:
    Min       1Q   Median       3Q      Max
-10.0014  -1.2286   0.0586   1.1802  13.8840

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -2.483610    0.813639  -3.052  0.00235 **
Age          -0.013359    0.022214  -0.601  0.54779
G            -0.014561    0.004993  -2.917  0.00365 **
GS           0.033034    0.006459   5.115 4.05e-07 ***
MP           0.357231    0.026967  13.247 < 2e-16 ***
X3P_perc     0.033674    0.006440   5.229 2.24e-07 ***
X2P_perc     0.026903    0.006783   3.966 8.03e-05 ***
FT_perc      0.008574    0.004350   1.971 0.04910 *
ORB          -0.531844    1.737325  -0.306 0.75960
DRB           0.712472    1.752494   0.407 0.68446
TRB          -0.153868    1.742248  -0.088 0.92965
AST           0.068270    0.113534   0.601 0.54782
STL          -0.467177    0.332059  -1.407 0.15989
BLK          -0.309885    0.326732  -0.948 0.34323
TOV          2.993232    0.248795  12.031 < 2e-16 ***
PF           -1.146724    0.191425  -5.990 3.33e-09 ***
PosPF        -0.352144    0.335708  -1.049 0.29455
PosPG        -1.188773    0.442684  -2.685 0.00741 **
PosSF        -0.160504    0.372835  -0.430 0.66697
PosSG        -0.367230    0.397801  -0.923 0.35624
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.308 on 710 degrees of freedom
Multiple R-squared:  0.8743,    Adjusted R-squared:  0.871
F-statistic: 260 on 19 and 710 DF,  p-value: < 2.2e-16
```

Figure 6: Residuals vs Fitted (Untransformed model).

4.2 Log-Transformed Model ($\log(\text{PTS} + 1)$)

- Multiple R^2 : 0.9095

- Adjusted R^2 : 0.9071
- Residual standard error: 0.2173 on 710 degrees of freedom
- Significant predictors: G, GS, MP, X3P_perc, X2P_perc, FT_perc, TOV

```
Call:
lm(formula = log(PTS + 1) ~ Age + G + GS + MP + X3P_perc + X2P_perc +
    FT_perc + ORB + DRB + TRB + AST + STL + BLK + TOV + PF +
    Pos, data = player_data_cleaned)

Residuals:
    Min       1Q   Median       3Q      Max
-0.70262 -0.13428  0.00037  0.14144  0.84694

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.1000125  0.0765962   1.306  0.19207
Age          -0.0009617  0.0020912  -0.460  0.64575
G             0.0014900  0.0004700   3.170  0.00159 **
GS           -0.0027088  0.0006080  -4.455  9.74e-06 ***
MP            0.0536509  0.0025386  21.134 < 2e-16 ***
X3P_perc     0.0072884  0.0006062  12.022 < 2e-16 ***
X2P_perc     0.0060632  0.0006385   9.496 < 2e-16 ***
FT_perc      0.0032417  0.0004095   7.917  9.41e-15 ***
ORB          -0.0653472  0.1635523  -0.400  0.68961
DRB          -0.0724324  0.1649802  -0.439  0.66077
TRB           0.0953824  0.1640157   0.582  0.56106
AST          -0.0174417  0.0106881  -1.632  0.10315
STL          -0.0201519  0.0312602  -0.645  0.51936
BLK          -0.0334214  0.0307586  -1.087  0.27760
TOV           0.1956303  0.0234216   8.353  3.50e-16 ***
PF           -0.0259062  0.0180208  -1.438  0.15100
PosPF        -0.0455965  0.0316036  -1.443  0.14953
PosFG        -0.0650669  0.0416744  -1.561  0.11889
PosSF        -0.0464634  0.0350987  -1.324  0.18600
PosSG        -0.0353289  0.0374491  -0.943  0.34580
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2173 on 710 degrees of freedom
Multiple R-squared:  0.9095,    Adjusted R-squared:  0.9071
F-statistic: 375.6 on 19 and 710 DF,  p-value: < 2.2e-16
```

Figure 7: Residuals vs Fitted (Log-transformed model).

Interpretation: Log transformation improved homoscedasticity and model fit (higher R^2 and reduced residual spread).

5. Diagnostic Checks

The residual plot of the original model with response PTS displays that as the values of the predictors increase, the dispersion among residuals increases, so there is noticeable heteroscedasticity in the error components. However, the residuals are distributed around 0, indicating that the expectation of the errors is 0.

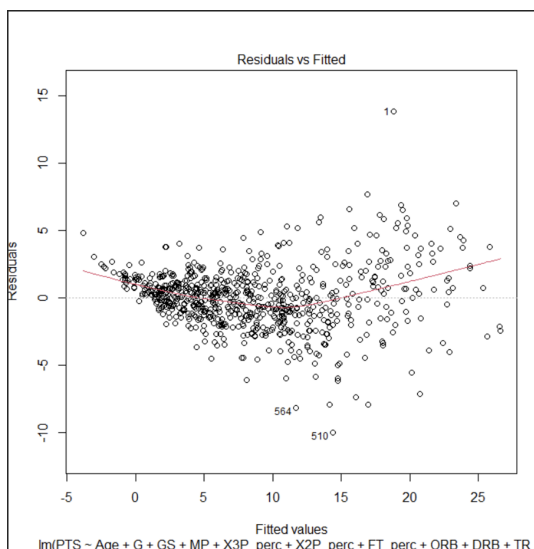


Figure 8: Zoomed-in Residuals vs Fitted (Log model) showing labeled notable observations.

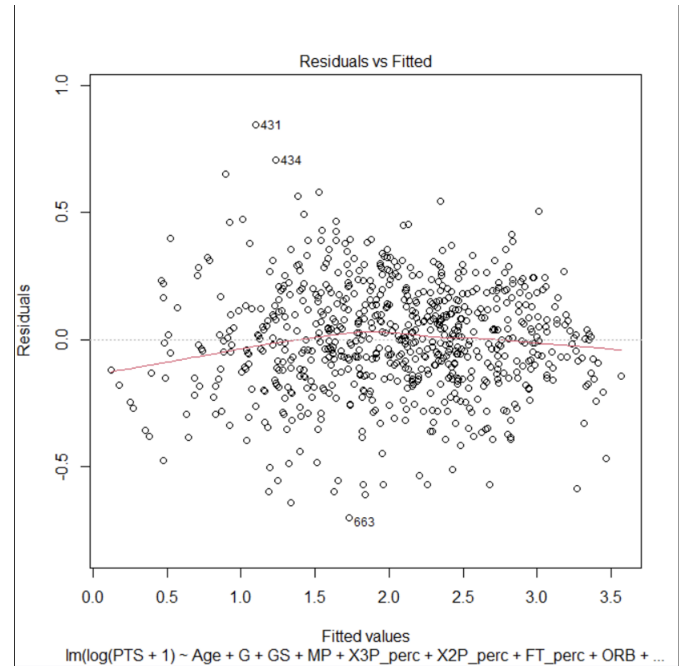


Figure 9: Q-Q plot of standardized residuals for the log-transformed model. Slight tail deviations but good central agreement with normality.

Here, we observe that the residuals are distributed more randomly, indicating Homoscedasticity. We also check for the normality of error terms using Q-Q (quantile-quantile) plot.

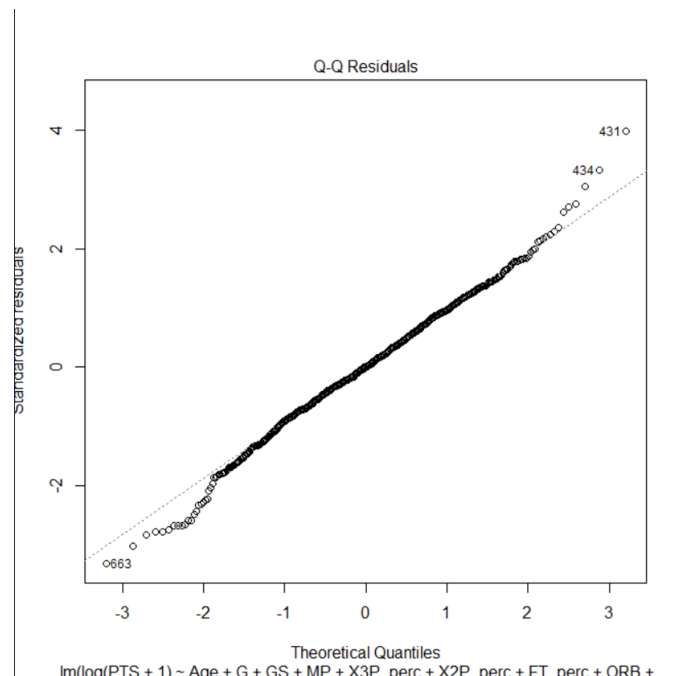


Figure 10: Cook's distance for all observations. Observations with largest influence are labeled.

From the plot, we can see there are some outliers to the tails of the plot, but overall the residuals follow normality. Finally, we check for the presence of outliers/leverage points using the cook's d measure. We get the following plot:-

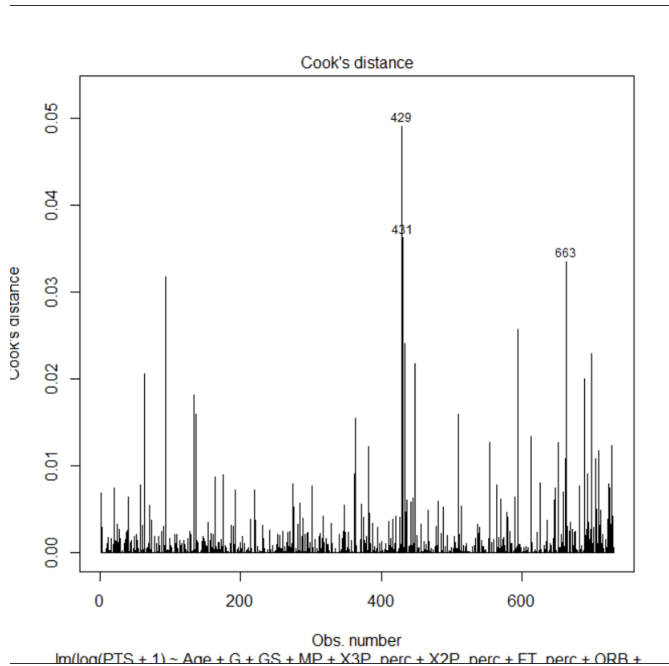


Figure 11: Cook's distance (alternate view) highlighting influential observations around indices of interest.

Table 1: Model assumptions, tests and results

Assumption	Test / Method	Result
Linearity	Residuals vs Fitted	Mostly linear. Residuals are centred around 0 and show an approximate linear relationship for most fitted values.
Normality of residuals	Q-Q plot & Shapiro-Wilk	Some tail deviations; Shapiro-Wilk $W = 0.966$, $p < 0.05$, indicating slight departure from perfect normality in the tails.
Homoscedasticity	Breusch-Pagan test	$p < 0.05$ for the raw model (heteroscedasticity present); variance stabilised after $\log(\text{PTS} + 1)$ transform.
Uncorrelated errors	Durbin-Watson statistic	DW ≈ 1.82 — indicates little to no autocorrelation of residuals.
Multicollinearity	Variance Inflation Factor (VIF)	Moderate VIF for X3P% and X2P% ; slight multicollinearity observed among rebound variables (ORB , DRB , TRB).
Outliers / Influential points	Cook's distance	Approximately 25 observations flagged as influential; overall influence not extreme but merits case-level review.

6. Interpretation of Key Predictors

- **Minutes played (MP):** Strongest positive effect on points — more minutes means more scoring opportunity.
- **Shooting efficiency:** **X3P_perc**, **X2P_perc**, and **FT_perc** positively influence PTS.
- **Turnovers (TOV):** Positively correlated in model — likely a proxy for usage rate (high-usage players both turn over and score more).
- **Games started / Games played:** Mixed effects; after controlling for MP, **GS** had a slight negative effect in the full model.
- **Position:** Point guards score slightly less than centers (baseline) after adjusting for other variables.

7. Limitations

- Multicollinearity among rebound statistics (**ORB**, **DRB**, **TRB**) inflates variance of coefficient estimates.
- Some outliers/influential points may bias parameter estimates.
- Only linear relationships were considered; interactions and non-linearities were not explored.
- Alternative approaches (Ridge/Lasso, WLS, or tree-based models) could improve predictive performance and handle multicollinearity.

8. Conclusion

The log-transformed linear model explains roughly 90–91% of variance in PTS. Primary drivers of scoring are minutes played, shooting percentages, turnovers (as a usage proxy), and games played. The model is a strong baseline

for prediction; future work should evaluate regularization, interaction terms, and advanced metrics for further gains.

Reproducibility

See the README.md file.

Project Contribution

- **Web Scraping:** Kuldip Manvar
- **Exploratory Data Analysis and Linear Re-**

gression Model: Pratya Bhukta, Kuldip Manvar, Manya Gupta, Annadevara Y. Chary

- **Shiny App:** Kuldip Manvar
- **Report Preparation and Documentation:** Kuldip Manvar, Pratya Bhukta

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